

# Paul Jerabek

Group Leader Computational Materials Design

Deputy Head of Department of Materials Design

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Email [paul.jerabek@hereon.de](mailto:paul.jerabek@hereon.de)

ORCID [0000-0002-2995-9755](https://orcid.org/0000-0002-2995-9755)

Scholar [profile](#)

LinkedIn [paul-jerabek](#)

Helmholtz-Zentrum Hereon, Max-Planck-Straße 1, 21502 Geesthacht

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## Education

- Dec 2010 – Sep 2014 **Dr. rer. nat.**, *Philipps-Universität Marburg*  
Thesis: *Clusterartige und hochkoordinierte Metallkomplexe*
- Jun 2010 – Nov 2010 **Dipl.-Chem.**, *Philipps-Universität Marburg*  
Thesis: *Bindungstheoretische Untersuchungen der Systeme des Typs YE-X-EY*
- Oct 2005 – Jun 2010 **Studies of Chemistry**, *Philipps-Universität Marburg*

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## Professional experience

- Sep 2022 – present **Deputy Head of Department of Materials Design**, *Helmholtz-Zentrum Hereon*  
Department administration and representation, third-party funding acquisition, research strategy planning, digital transformation.
- Oct 2021 – present **Co-Leader, German-New Zealand Green Hydrogen Centre (He Honoka Hauwai)**, *Helmholtz-Zentrum Hereon / University of Otago*  
Administration and representation, strategic planning, networking, funding acquisition, event organization.
- Sep 2019 – present **Group Leader “Metal Hydride Modeling”**, *Helmholtz-Zentrum Hereon*  
First-principles molecular dynamics, CALPHAD and phase-field simulations for the characterization and optimization of novel metal hydride materials for large-scale hydrogen storage. Close collaborations with Lawrence Livermore National Laboratory (USA) and the University of Otago (New Zealand).
- Jul 2018 – Aug 2019 **Humboldt Return Fellow**, *Max-Planck-Institut für Kohlenforschung*  
Theoretical investigation of physical properties of organesson; substituent and solvent effects on switching temperature in Fe(II) spin-crossover complexes; reaction mechanism exploration for asymmetrical, homogeneous catalysis; bonding situation and electronic properties of pincer complexes.
- Dec 2017 – Jun 2018 **Postdoctoral Research Assistant**, *Massey University*  
Monte-Carlo melting simulations of heavy and superheavy elements; first-principles molecular dynamics simulations of coinage metals in liquid and solid phase; highly accurate calculations of atomic polarizabilities.
- Dec 2015 – Nov 2017 **Feodor Lynen Research Fellow**, *Massey University*  
Photochemistry of small molecules in different environments; influence of relativistic and solvent effects on atomic and molecular bonding properties; development of highly accurate ab-initio many-body potentials for heavy and superheavy elements.
- Dec 2014 – Nov 2015 **Postdoctoral Research Assistant**, *Philipps-Universität Marburg*  
Method development (C) for core electron spectroscopy of organic molecules on surfaces; density functional theory based investigation of transition metal clusters and complexes.

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| Dec 2010 – Nov 2014 | <b>Research Assistant, <i>Philipps-Universität Marburg</i></b><br>Quantum chemical investigation of main-group and transition metal compounds in collaboration with experimental groups in Bochum, München, Göttingen, Dresden, Münster and San Diego. Redesign of the lab course in Theoretical Chemistry. |
| Jan 2007 – Dec 2010 | <b>Student Assistant, <i>Philipps-Universität Marburg</i></b><br>Supervision of lab courses in Physical Chemistry and Chemistry for Medical Students.                                                                                                                                                       |

## Third-party funding

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| 2024 | <b>Sustainable TiFe Generation from Secondary Sources</b><br>BMBF (Germany)   Principal Investigator   1.5 M€   3 years                                                                   |
| 2024 | <b>HIDA Visiting Researcher Grant (with S. Faouri)</b><br>Helmholtz Association   Host   10 k€   2 months                                                                                 |
| 2023 | <b>HIDA Visiting Researcher Grant (with A. Rowberg)</b><br>Helmholtz Association   Host   15 k€   3 months                                                                                |
| 2022 | <b>Generating Hydrogen Storage Materials from New Zealand Resources</b><br>BMBF (Germany)   Coordinator   400 k€   3 years                                                                |
| 2021 | <b>Establishing a German-New Zealand Research Centre for Green Hydrogen</b><br>BMBF (Germany)   Coordinator   760 k€   5 years                                                            |
| 2021 | <b>HIDA Visiting Scientist Grant (with M. Selzer)</b><br>Helmholtz Association   Fellow   10 k€   3 months                                                                                |
| 2017 | <b>Humboldt Return Fellowship</b><br>Alexander von Humboldt Foundation   Fellow   47 k€   1 year                                                                                          |
| 2016 | <b>Computation of highly accurate interatomic potentials of heavy and superheavy elements</b><br>New Zealand eScience Infrastructure (NeSI)   Principal Investigator   117 k€   1.5 years |
| 2014 | <b>Hochpräzise quantenchemische Simulationen von atomaren und molekularen Phasen unter extremen Bedingungen</b><br>Alexander von Humboldt Foundation   Fellow   85 k€   2 years           |

## Conference organization

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| 2025 | <b>E-MRS Fall Meeting 2025 — Symposium A</b><br>Co-organizer, Warsaw, Poland                                                                             |
| 2024 | <b>E-MRS Fall Meeting 2024 — Symposium C: Sustainable materials for chemical and electrochemical energy storage II</b><br>Main organizer, Warsaw, Poland |
| 2023 | <b>Inaugural New Zealand Hydrogen Symposium</b><br>Co-organizer, Dunedin, New Zealand                                                                    |
| 2015 | <b>Bonding Analysis Workshop</b><br>Conceptualization and main organizer, Marburg, Germany                                                               |

## Teaching

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| 2025 – present | <b>Hydrogen Technology</b><br>Co-lecturer, Technische Universität Hamburg (TUHH), Hamburg |
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- 2019 **Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa (Volkswagen-Stiftung)**  
Tutor, University of Mauritius, Reduit
- 2014 – 2015 **Lectureship**  
Lecturer, Technische Hochschule Mittelhessen, Gießen
- 2014 – 2019 **Hands-on workshops on electronic structure methods**  
Workshop lecturer, Various universities, Hyderabad, Dunedin, Denton, Mariapfarr, Mauritius
- 2010 – 2014 **Practical course in Theoretical Chemistry**  
Teaching assistant, Philipps-Universität Marburg, Marburg
- 2007 – 2010 **Physical Chemistry and Chemistry for Medical Students**  
Student assistant, Philipps-Universität Marburg, Marburg

## Talks and posters

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- May 2025 **German-New Zealand WTZ Meeting 2025**, Wellington, New Zealand  
Invited Talk
- Feb 2025 **11th International Conference on Advanced Materials and Nanotechnology (AMN11)**, Christchurch, New Zealand  
Invited Talk
- Jul 2024 **New Zealand-Germany Energy Dialogue 2024**, Berlin, Wellington, Germany, New Zealand  
Invited Talk
- Feb 2023 **10th International Conference on Advanced Materials and Nanotechnology (AMN10)**, Rotorua, New Zealand  
Invited Talk
- Jul 2018 **18th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2018)**, Rota, Spain  
Invited Talk
- Jul 2016 **16th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2016)**, Rota, Spain  
Invited Talk
- Mar 2024 **German Physical Society (DPG) Spring Meeting 2024**, Berlin, Germany  
Contributed Talk
- Jan 2024 **2nd New Zealand Hydrogen Symposium (NZHS-2)**, Wellington, New Zealand  
Contributed Talk
- Mar 2023 **German Physical Society (DPG) Spring Meeting 2023**, Dresden, Germany  
Contributed Talk
- Sep 2022 **European Materials Research Society (E-MRS) Fall Meeting 2022**, Warsaw, Poland  
Contributed Talk
- Mar 2019 **Chemiedozententagung 2019**, Koblenz, Germany  
Contributed Talk
- Nov 2017 **17th Biennial Symposium of the Australian and New Zealand Associations of von Humboldt Fellows**, Wellington, New Zealand  
Contributed Talk
- Sep 2017 **12th International Conference on Relativistic Effects in Heavy-Element Chemistry and Physics (REHE)**, Marburg, Germany  
Contributed Talk

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| Aug 2016 | <b>New Zealand Institute of Chemistry Conference (NZIC-16)</b> ,<br>Queenstown, New Zealand<br>Contributed Talk                |
| May 2024 | <b>18th International Symposium of Metal-Hydrogen Systems (MH 2024)</b> , Saint-Malo, France<br>Poster                         |
| Sep 2018 | <b>54th Symposium on Theoretical Chemistry (STC)</b> , Halle (Saale),<br>Germany<br>Poster                                     |
| Aug 2017 | <b>11th World Association of Theoretical and Computational Chemists (WATOC)</b> , Munich, Germany<br>Poster                    |
| Jul 2016 | <b>9th Congress of the International Society for Theoretical Chemical Physics (ISTCP IX)</b> , Grand Forks (ND), USA<br>Poster |
| Sep 2015 | <b>51st Symposium on Theoretical Chemistry (STC)</b> , Postdam, Germany<br>Poster                                              |
| Sep 2013 | <b>49th Symposium on Theoretical Chemistry (STC)</b> , Erlangen, Germany<br>Poster                                             |
| Sep 2012 | <b>48th Symposium on Theoretical Chemistry (STC)</b> , Karlsruhe, Germany<br>Poster                                            |
| Oct 2011 | <b>Modeling of Molecular Properties Conference (MolMod)</b> , Heidelberg, Germany<br>Poster                                    |
| Jul 2011 | <b>9th World Association of Theoretical and Computational Chemists (WATOC)</b> , Santiago de Compostela, Spain<br>Poster       |
| Sep 2010 | <b>46th Symposium on Theoretical Chemistry (STC)</b> , Münster, Germany<br>Poster                                              |
| Feb 2025 | <b>University of Waikato</b> , Hamilton, New Zealand<br>Invited Seminar                                                        |
| Jun 2024 | <b>University of Hamburg</b> , Hamburg, Germany<br>Invited Seminar                                                             |
| Feb 2024 | <b>Massey University</b> , Auckland, New Zealand<br>Invited Seminar                                                            |
| Feb 2024 | <b>University of Waikato</b> , Hamilton, New Zealand<br>Invited Seminar                                                        |
| Mar 2023 | <b>Massey University</b> , Auckland, New Zealand<br>Invited Seminar                                                            |
| Mar 2023 | <b>University of Auckland</b> , Auckland, New Zealand<br>Invited Seminar                                                       |
| Dec 2021 | <b>Karlsruhe Institute of Technology (KIT)</b> , Karlsruhe, Germany<br>Invited Seminar                                         |
| May 2019 | <b>Helmholtz-Zentrum für Material- und Küstenforschung</b> , Geesthacht, Germany<br>Invited Seminar                            |
| Apr 2019 | <b>Universität Graz</b> , Graz, Austria<br>Invited Seminar                                                                     |
| Feb 2019 | <b>Technische Universität Berlin</b> , Berlin, Germany<br>Invited Seminar                                                      |

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| Jan 2019 | <b>GSI Helmholtzzentrum für Schwerionenforschung</b> , Darmstadt, Germany<br>Invited Seminar                                                                                    |
| Feb 2018 | <b>University of Otago</b> , Dunedin, New Zealand<br>Invited Seminar                                                                                                            |
| Jun 2017 | <b>University of Otago</b> , Dunedin, New Zealand<br>Invited Seminar                                                                                                            |
| Jul 2016 | <b>University of North Texas</b> , Denton (TX), USA<br>Invited Seminar                                                                                                          |
| Oct 2019 | <b>Summer School/Workshop on Computational Chemistry “Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa”</b> , Reduit, Mauritius<br>Workshop Lecture |
| Mar 2019 | <b>XIIth Workshop on Modern Methods in Quantum Chemistry</b> , Mariapfarr, Austria<br>Workshop Lecture                                                                          |
| Dec 2017 | <b>University of Otago</b> , Dunedin, New Zealand<br>Workshop Lecture                                                                                                           |
| Jul 2016 | <b>University of North Texas</b> , Denton (TX), USA<br>Workshop Lecture                                                                                                         |
| Jan 2014 | <b>MCBR Winter School</b> , Hyderabad, India<br>Workshop Lecture                                                                                                                |

## Peer-reviewed journal articles

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70. E. Pericoli, V. Ferretti, E. Alvares, **P. Jerabek**, C. Pistidda, L. Pasquini, Ni–Cr Substitution in TiFe: An Integrated Experimental–Modeling Study of Phase Stability and Hydride Thermodynamics, *Adv. Energy Sustain. Res.* **2026**, 7, e70243.
69. J.-E. Ahrens, **P. Jerabek**, L. Vondung, Relativistic effects are crucial for uranium-phosphorus bonding: a DFT study of cyclopentadienyluranium-phosphine complexes, *Mol. Phys.* **2026**, e2611400.
68. A. J. E. Rowberg, S. Kang, K. Sellschopp, **P. Jerabek**, T. W. Heo, B. C. Wood, Role of Elemental Substitution on Hydrogen Incorporation in Characteristic Ti/Fe-Based Oxides, *ACS Appl. Energy Mater.* **2026**, 9, 1242–1253.
67. Y. Shang, A. Santhosh, **P. Jerabek**, T. Klassen, C. Pistidda, Leveraging atomic disorder to modulate hydrogen storage thermodynamics in compositionally complex intermetallics, *Interdiscip. Mater.* **2026**, 0, 1–13.
66. L. Dinachandran, E. Alvares, K. Sellschopp, T. Klassen, **P. Jerabek**, A. L. Garden, Predicting hydrogen storage capacity loss of FeTi by ab initio thermodynamic modelling of Fe–Ti–Si phase equilibria, *Acta Mater.* **2025**, 301, 121514.
65. A. Chandran, A. Santhosh, **P. Jerabek**, R. C. Aydin, C. J. Cyron, TiAlNb alloy interatomic potentials: comparing passive and active machine learning techniques with MTP and DeePMD, *Front. Mater.* **2025**, 12.
64. Y. Shang, A. Santhosh, M. Ante, L. Wimbert, **P. Jerabek**, T. Schupp, T. Klassen, C. Pistidda, Low-temperature regeneration of metal hydride after impurity gas exposure, *Acta Mater.* **2025**, 298, 121398.
63. T. T. Le, K. Sellschopp, F. Murgia, A. L. Garden, S. Bordignon, J. P. Embs, M. R. Chierotti, A. Schökel, F. Karimi, **P. Jerabek**, T. Klassen, C. Pistidda, High Ionic Conduction in Rb-

- and Cs-Mixed Cation Amide for Energy Storage, *Small* **2025**, 2502943.
62. E. Alvares, A. J. E. Rowberg, K. Sellschopp, B. C. Wood, T. Klassen, **P. Jerabek**, C. Pistidda, A Thermodynamic Approach to Model Multicomponent FeTi-based Alloys for Hydrogen Storage, *Scr. Mater.* **2025**, 259, 116516.
  61. E. Alvares, K. Sellschopp, B. Wang, S. Kang, T. Klassen, T. W. Heo, T. Klassen, **P. Jerabek**, C. Pistidda, Multiscale modeling of metal-hydride interphases—quantification of decoupled chemo-mechanical energies, *npj Comput. Mater.* **2024**, 10, 249.
  60. A. Chandran, A. Santhosh, C. Pistidda, **P. Jerabek**, R. Aydin, C. J. Cyron, Comparative analysis of ternary TiAlNb interatomic potentials: moment tensor vs. deep learning approaches, *Front. Mater.* **2024**, 11, 1466793.
  59. P. Hannappel, E. Alvares, F. Heubner, C. Pistidda, **P. Jerabek**, T. Weißgärber, Thermodynamic assessment of the Ce-H and CeNi<sub>5</sub>-H system, *Calphad* **2024**, 85, 102701.
  58. M. Robb, L. Bondi, S. Rodriguez-Jimenez, A. Garden, **P. Jerabek**, S. Brooker, Predictable electronic tuning of FeII and RuII complexes via choice of azine: correlation of ligand pK<sub>a</sub> with E<sub>pa</sub>(MIII/II) of complex, *Dalton Trans.* **2024**, 53, 1999-2007.
  57. A. Robles-Navarro, **P. Jerabek**, P. Schwerdtfeger, Tipping the balance between the bcc and fcc phase within the alkali and coinage metal groups, *Angew. Chem. Int. Ed.* **2024**, 63, e202313679.
  56. Y. Shang, A. Santhosh, **P. Jerabek**, T. Klassen, C. Pistidda, First-principles study on interfacial property in MgB<sub>2</sub>-based reactive hydride composites, *Scr. Mater.* **2024**, 240, 115837.
  55. M. Ghazanfari, K. Siemensmeyer, A. Santhosh, J. Vrijmoed, M. Tallu, S. Dehnen, **P. Jerabek**, G. Thiele, Low-Cost, Multifunctional, and Sustainable Sodium Sulfido Ferrate(II), *Inorg. Chem.* **2023**, 62, 15358-15366.
  54. Y. Shang, Z. Lei, E. Alvares, S. Garroni, R. Dore, M. Rustici, S. Enzo, A. Schökel, Y. Shi, **P. Jerabek**, Z. Lu, T. Klassen, C. Pistidda, Ultra-lightweight compositionally complex alloys with large ambient-temperature hydrogen storage capacity, *Mater. Today* **2023**, 67, 113-126.
  53. A. Santhosh, S. Kang, N. Keilbart, B. C. Wood, T. Klassen, **P. Jerabek**, M. Dornheim, Influence of near-surface oxide layers on TiFe hydrogenation - Mechanistic insights and implications for hydrogen storage applications, *J. Mater. Chem. A* **2023**, 11, 18776-18789.
  52. T.-T. Le, A. Santhosh, S. Bordignon, M. R. Chierotti, **P. Jerabek**, T. Klassen, C. Pistidda, Experimental and Computational Studies on the Formation of Mixed Amide-Hydride Solid Solutions for CsNH<sub>2</sub>-CsH System, *Results in Engineering* **2023**, 17, 100895.
  51. L. Pasquini, K. Sakaki, E. Akiba, **P. Jerabek** et al., Magnesium- and intermetallic alloys-based hydrides for energy storage: modelling, synthesis and properties, *Prog. Energy* **2022**, 4, 32007.
  50. **P. Jerabek**, A. Santhosh, P. Schwerdtfeger, Relativistic Effects Stabilize Unusual Gold(II) Sulfate Structure via Aurophilic Interactions, *Inorg. Chem.* **2022**, 61, 13077-13084.
  49. **P. Jerabek**, A. Burrows, P. Schwerdtfeger, Solving a problem with a single parameter: A smooth bcc to fcc phase transition for metallic lithium, *Chem. Commun.* **2022**, 58, 13369-13372.
  48. M. R. Ghazanfari, L. Vittadello, D. Al-Sabbagh, A. Santhosh, C. Frankcom, F. Fuß, C. A. von Randow, K. Siemensmeyer, J. C. Vrijmoed, F. Emmerling, **P. Jerabek**, M. Imlau, G. Thiele,

Remarkable Infrared Nonlinear Optical, Dielectric, and Strong Diamagnetic Characteristics of Semiconducting K<sub>3</sub>[BiS<sub>3</sub>], *J. Phys. Chem. Lett.* **2022**, *13*, 6987–6993.

47. M. R. Ghazanfari, A. Santhosh, J. C. Vrijmoed, K. Siemensmeyer, B. Peters, S. Dehnen, **P. Jerabek**, G. Thiele, Large-scale synthesis of mixed valance K<sub>3</sub>[Fe<sub>2</sub>S<sub>4</sub>] with high dielectric and ferrimagnetic characteristics, *RSC Adv.* **2022**, *12*, 30514–30521.
46. M. R. Ghazanfari, A. Santhosh, K. Siemensmeyer, F. Fuß, L. Staab, J. C. Vrijmoed, B. Peters, M. Liesegang, S. Dehnen, O. Oeckler, **P. Jerabek**, G. Thiele, Large Exchange Bias, High Dielectric Constant, and Outstanding Ionic Conductivity in a Single-Phase Spin Glass, *Adv. Electron. Mater.* **2022**, *8*, 2200483.
45. E. Florez, O. R. Smits, J.-M. Mewes, **P. Jerabek**, P. Schwerdtfeger, From the gas phase to the solid state: The chemical bonding in the superheavy element flerovium, *J. Chem. Phys.* **2022**, *157*, 64304.
44. D. M. Dreistadt, L. Thi-Thu, G. Capurso, J. M. Bellosta von Colbe, A. Santhosh, C. Pistidda, N. Scharnagl, H. Ovri, C. Milanese, **P. Jerabek**, T. Klassen, J. Jepsen, An Effective Activation Method for Industrially Produced TiFeMn Powder for Hydrogen Storage, *J. Alloys Compd.* **2022**, *919*, 165847.
43. L. Bondi, A. L. Garden, F. Totti, **P. Jerabek**, S. Brooker, Quantitative assessment of ligand substituent effects on  $\sigma$ - and  $\pi$ -contributions to Fe-N bonds in spin crossover FeII complexes, *Chem. Eur. J.* **2022**, *28*, e202104314.
42. E. Alvares, **P. Jerabek**, Y. Shang, A. Santhosh, C. Pistidda, T. W. Heo, B. Sundman, M. Dornheim, Modeling the thermodynamics of the FeTi hydrogenation under para-equilibrium: An ab-initio and experimental study, *Calphad* **2022**, *77*, 102426.
41. C. Pistidda, A. Santhosh, **P. Jerabek**, Y. Shang, A. Girella, C. Milanese, M. Dore, S. Garroni, S. Bordignon, M. R. Chierotti, T. Klassen, M. Dornheim, Hydrogenation via a low energy mechanochemical approach: The MgB<sub>2</sub> case, *J. Phys. Energy* **2021**, *3*, 44001.
40. O. R. Smits, J. Mewes, **P. Jerabek**, P. Schwerdtfeger, Oganesson: A Noble Gas Element That Is Neither Noble Nor a Gas, *Angew. Chem. Int. Ed.* **2020**, *59*, 23636–23640.
39. O. R. Smits, **P. Jerabek**, E. Pahl, P. Schwerdtfeger, First-principles melting of krypton and xenon based on many-body relativistic coupled-cluster interaction potentials, *Phys. Rev. B* **2020**, *101*, 104103.
38. M. Buchsteiner, L. Martinez-Rodriguez, **P. Jerabek**, I. Pozo, M. Patzer, N. Nöthling, C. W. Lehmann, A. Fürstner, Catalytic Asymmetric Fluorination of Copper Carbene Complexes: Preparative Advances and a Mechanistic Rationale, *Chem. Eur. J.* **2020**, *26*, 2509–2515.
37. L. Bondi, A. L. Garden, **P. Jerabek**, F. Totti, S. Brooker, Quantitative and Chemically Intuitive Evaluation of the Nature of M–L Bonds in Paramagnetic Compounds: Application of EDA-NOCV Theory to Spin Crossover Complexes, *Chem. Eur. J.* **2020**, *26*, 13677–13685.
36. L. Vondung, **P. Jerabek**, R. Langer, Ligands Based on Phosphine-Stabilized Aluminum(I), Boron(I), and Carbon(0), *Chem. Eur. J.* **2019**, *25*, 3068–3076.
35. J.-M. Mewes, **P. Jerabek**, O. R. Smits, P. Schwerdtfeger, Oganesson Is a Semiconductor: On the Relativistic Band-Gap Narrowing in the Heaviest Noble-Gas Solids, *Angew. Chem. Int. Ed.* **2019**, *58*, 14260–14264.
34. **P. Jerabek**, O. R. Smits, J.-M. Mewes, K. A. Peterson, P. Schwerdtfeger, Solid Oganesson via a Many-Body Interaction Expansion Based on Relativistic Coupled-Cluster Theory and

- from Plane-Wave Relativistic Density Functional Theory, *J. Phys. Chem. A* **2019**, *123*, 4201–4211.
33. **P. Jerabek**, P. Schwerdtfeger, G. Frenking, Dative and Electron-Sharing Bonding in Transition Metal Compounds, *J. Comput. Chem.* **2019**, *40*, 247–264.
  32. O. Smits, **P. Jerabek**, E. Pahl, P. Schwerdtfeger, A Hundred-Year-Old Experiment Re-evaluated: Accurate Ab-Initio Monte-Carlo Simulations of the Melting of Radon, *Angew. Chem. Int. Ed.* **2018**, *57*, 9961–9964.
  31. J.-M. Mewes, **P. Jerabek**, D. S. Bohle, P. Schwerdtfeger, The Light-Driven Isomerization of Aqueous Nitrate: A Theoretical Perspective, *ChemPhotoChem* **2018**, *2*, 725–733.
  30. **P. Jerabek**, L. Vondung, P. Schwerdtfeger, Tipping the Balance between Ligand and Metal Protonation due to Relativistic Effects: Unusually High Proton Affinity in Gold(I) Pincer Complexes, *Chem. Eur. J.* **2018**, *24*, 6047–6051.
  29. **P. Jerabek**, O. Smits, E. Pahl, P. Schwerdtfeger, A relativistic coupled-cluster interaction potential and rovibrational constants for the xenon dimer, *Mol. Phys.* **2018**, *116*, 1–8.
  28. **P. Jerabek**, P. Schwerdtfeger, J. K. Nagle, Static dipole polarizability of palladium from relativistic coupled-cluster theory, *Phys. Rev. A* **2018**, *98*, 12508.
  27. **P. Jerabek**, B. Schuetrumpf, P. Schwerdtfeger, W. Nazarewicz, Electron and nucleon localization functions of oganesson: Approaching the Thomas-Fermi limit, *Phys. Rev. Lett.* **2018**, *120*, 53001.
  26. J. Hornung, J. Weßing, **P. Jerabek**, C. Gemel, A. Pöthig, G. Frenking, R. A. Fischer, Suppressed Phosphine Dissociation by Polarization Effects on the Donor–Acceptor Bonds in  $[\text{Ni}(\text{PEt}_3)_{4-n}(\text{ECp}^*)_n]$  ( $E = \text{Al}, \text{Ga}$ ), *Inorg. Chem.* **2018**, *57*, 12657–12664.
  25. **P. Jerabek**, B. von der Esch, H. Schmidbaur, P. Schwerdtfeger, Influence of Relativistic Effects on Bonding Modes in M(II) Dinuclear Complexes ( $M = \text{Au}, \text{Ag}, \text{and Cu}$ ), *Inorg. Chem.* **2017**, *56*, 14624–14631.
  24. Q. Zhang, **P. Jerabek**, M. Chen, M. Zhou, G. Frenking, The Oxygen-Rich Beryllium Oxides  $\text{BeO}_4$  and  $\text{BeO}_6$ , *Angew. Chem. Int. Ed.* **2016**, *55*, 10863–10867.
  23. M. Klues, **P. Jerabek**, T. Breuer, M. Oehzelt, K. Hermann, R. Berger, G. Witte, Understanding the F 1s NEXAFS Dichroism in Fluorinated Organic Semiconductors, *J. Phys. Chem. C* **2016**, *120*, 12693–12705.
  22. K. Freitag, M. Molon, **P. Jerabek**, K. Dilchert, C. Rösler, R. W. Seidel, C. Gemel, G. Frenking, R. A. Fischer, Zn...Zn interactions at nickel and palladium centers, *Chem. Sci.* **2016**, *7*, 6413–6421.
  21. K. Freitag, C. Gemel, **P. Jerabek**, M. I. Oppel, R. W. Seidel, G. Frenking, H. Banh, K. Dilchert, R. A. Fischer, The  $\sigma$ -Aromatic Clusters  $[\text{Zn}_3]^+$  and  $[\text{Zn}_2\text{Cu}]$ : Embryonic Brass, *Angew. Chem. Int. Ed.* **2015**, *54*, 4370–4374.
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